Yiting Zhang

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Address: Dalian University of Technology, Dalian, Liaoning, China

Education

Dalian University of Technology (DUT, Project 985 & 211, Double First-Class University) **09/2023–06/2027**

Bachelor of Engineering in **Software Engineering**

Overall GPA: 4.2/5; 92.0/100 Rank: 13/398

Core Courses: C++ Programming (99), Probability and Statistics (98), Compiler Technology (98), Software Engineering (97), C Programming (97), Database Systems (96), Linear Algebra and Analytic Geometry (96), Discrete Mathematics (96), College Physics (95), Engineering Mathematical Analysis I (95)

Publications

Yiting Zhang, Zhixiang Xu, Xiaoya Fan, and Zhong Wang. "SCBNet: A Safety-Driven Class-Adaptive Boundary-Aware Network for Semantic Segmentation in Foggy Autonomous Driving Scenes." 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2026.

Yiting Zhang and S. Huang. "Dynamic Time Series Prediction with Deep Fusion of ARIMA and TimeXer Models." Mechatronics, Control and AI (MCANS), 2025. (EI Compendex)

Yiting Zhang*, Wenkun Sun, Shing-Tung Yau, Zhong Wang, et al. "Estimating Unknown Parameters in Differential Equations with a Reinforcement Learning-Based PSO Method." Manuscript in preparation.

Research Experience

SCBNet: A Safety-Driven Class-Adaptive Boundary-Aware Network for Semantic Segmentation in Foggy Autonomous Driving Scenes **01/2025–07/2026**

Research Assistant and Project Lead Advisor: Prof. Zhong Wang BIMSA

- Proposed the Parallel Attention Enhancement Module (PAEM): To address layer-by-layer error amplification and boundary-feature degradation caused by conventional sequential attention in adverse weather such as fog, computed two attention mechanisms in parallel and fused them adaptively, removing the serial information bottleneck and enabling more stable recovery of fog-obscured boundary details;
- Designed safety-driven boundary supervision: To reduce boundary-recognition errors for safety-critical objects such as pedestrians and vehicles and resolve inconsistent boundaries across branch outputs in foggy scenes, jointly applied CAELoss (Class-Adaptive Edge Loss) and BCLoss (Boundary Consistency Loss). CAELoss assigns greater training weight to critical-class boundaries to improve segmentation accuracy, while BCLoss aligns PAEM's parallel-branch outputs to ensure consistent final boundary predictions;
- Built the complete experimental pipeline independently in Python using PyTorch and DeepLabV3+ as the base architecture, covering data preprocessing, model training, and metric computation; designed and conducted ablation studies, comparisons with state-of-the-art models in foggy scenes, and cross-weather generalization experiments involving rain, snow, and nighttime conditions;
- Results: Achieved 78.35% mIoU, 86.16% mAcc, and 96.17% aAcc on the Foggy Cityscapes validation set, ranking first among all compared methods on all three metrics; outperformed leading state-of-the-art models including VMamba, Mask2Former, and SegMAN; and demonstrated zero-shot generalization on ACDC-Rain/Snow/Night and Foggy Zurich;
- Independently completed the full manuscript-writing and submission process as first author; the paper was accepted as a Regular Paper at IEEE SMC 2026.

Estimating Unknown Parameters in Differential Equations with a Reinforcement Learning-Based PSO Method **01/2024–Present**

Research Assistant **Advisor: Prof. Zhong Wang BIMSA**

Open-source code: https://github.com/zw427/DERLPSO/tree/restructure

- Designed and implemented the core algorithms for the stochastic differential equation (SDE) module, independently developed it in Python. Key contributions include:
  - Designed a decoupled solution strategy for stochastic and deterministic parameters: Proposed a two-stage "grid search + particle swarm optimization (PSO)" framework that enumerates candidate stochastic-noise parameters by grid search and passes each candidate set to PSO as fixed values, allowing PSO to focus on deterministic parameters and reducing stochastic interference in optimization;
  - Designed a statistical-test-based residual filtering mechanism: Introduced a Shapiro–Wilk-test-based penalty for residual normality into the fitness function, making parameter estimates more consistent with the statistical assumptions of SDEs;
  - Independently implemented numerical solutions for multidimensional SDEs using the Euler–Maruyama method and introduced Cholesky decomposition to model correlations among multidimensional Brownian motions, enabling the model to handle complex systems with correlated noise across variables;
  - Validated the proposed hybrid RL-PSO method on several standard dynamical systems, including geometric Brownian motion and the Ornstein–Uhlenbeck process;
- Manuscript in preparation for submission to Engineering Applications of Artificial Intelligence (SCI Q1).

MedSeg AI—A FastAPI-Based Web Platform for Ultrasound Tumor Segmentation **12/2025–06/2026**
Project Lead **Advisor: Prof. Yonggang Fan DUT**

- Independently implemented seven mainstream semantic segmentation models in PyTorch—vanilla UNet, UNet with pretrained ResNet50/VGG16 backbones, UNet++, Attention UNet, R2UNet, and FCN—and adapted them for breast-ultrasound image segmentation; designed a unified training, testing, and batch-evaluation framework supporting rapid model switching and large-scale comparative experiments;
- Implemented a comprehensive segmentation evaluation system covering standard medical-image metrics, including precision, recall, mIoU (mean intersection over union), mPA (mean pixel accuracy), and the Dice coefficient, and generated visualizations such as confusion matrices;
- Refactored the original PyQt desktop application into a FastAPI web service, enabling online inference with multiple models, segmentation visualization, and automatic generation of PDF diagnostic reports;
- Results: Entered the 2026 China Collegiate Computing Contest—Network Technology Challenge, won First Prize in the Northeast Regional Division, and advanced to the national finals (results pending).

World Engine — Research Intern, Multi-Camera Model **09/2026–Present**
Research Assistant **Advisor: Prof. Yue Wang USC**

- Working on large-scale multi-camera human and hand pose estimation for embodied AI and video understanding, focusing on data processing, pose reconstruction, and evaluation across synchronized multi-view videos;
- Building and validating an end-to-end data pipeline integrating hand detection, 3D hand reconstruction, and multi-view processing, including WiLoR- and HaMeR-based components for large-scale video datasets;
- Conducting systematic data-quality inspection and failure-case analysis for hand/whole-body pose estimation, identifying detection, reconstruction, and cross-view consistency issues to improve training-data reliability;
- Collaborating with the research team on preparing large-scale pose and video data for subsequent embodied perception and vision-language-action (VLA) model development.

Honors and Awards

- **National Scholarship, Ministry of Education of China (Top 0.2%) (2025, 2026)**
- First Prize Scholarship for Academic Excellence, Dalian University of Technology (Top 5%) (2024, 2025)
- Scholarship for Innovation in Science and Technology, Dalian University of Technology (Top 4%) (2025)
- Scholarship for Excellence in Cultural and Sports Activities, Dalian University of Technology (Top 6%) (2026)
- Mitacs Globalink Research Internship (Fully Funded), Mitacs Canada, 2026
- Meritorious Winner, COMAP Mathematical Contest in Modeling (MCM/ICM) (Top 6%) (2025)
- Bronze Award, 5th Singapore Art Design Contest (SGADC), AIGC Track (2025)
- Second Prize, Northeast China Mathematical Modeling Contest (Provincial, Top 5%–10%) (2025)
- Second Prize, MathorCup Mathematical Application Challenge (Provincial, Top 15%) (2025)
- Third Prize (Provincial), China Collegiate Computing Contest—AIGC Innovation Contest (2025)
- Second Prize (Provincial), China Undergraduate Mathematical Contest in Modeling (CUMCM) (2025)
- Third Prize (National), National College Student Advertising Art Competition (2025)
- Third Prize (Provincial), National College Computer Ability Challenge (2024)
- Second Prize (Provincial), 6th National College Computer Ability Challenge (NCCAC) (2024)

Language Proficiencies

IELTS: 7.0 (Listening: 8.0; Speaking: 6.0; Reading: 7.5; Writing: 6.0) **04/2026**
CET-6: 551 **02/2025**

Professional Skills

Proficient in Python, C++, and Java; familiar with R, Rust, and other programming languages
Experienced with Git/GitHub for version control; proficient in the PyTorch deep learning framework

Software Copyrights

Autonomous Driving Safety Strategy Execution System V1.0 (Registration No. 2025SR2390892)
High-Precision Path Planning Software for Autonomous Driving V1.0 (Registration No. 2025SR2388363)
Multimodal Semantic Understanding and Analysis Software V1.0 (Registration No. 2026SR0133218)
Intelligent Interpretation System for Computer-Aided Medical Image Diagnosis V1.0 (Registration No. 2025SR1852612)